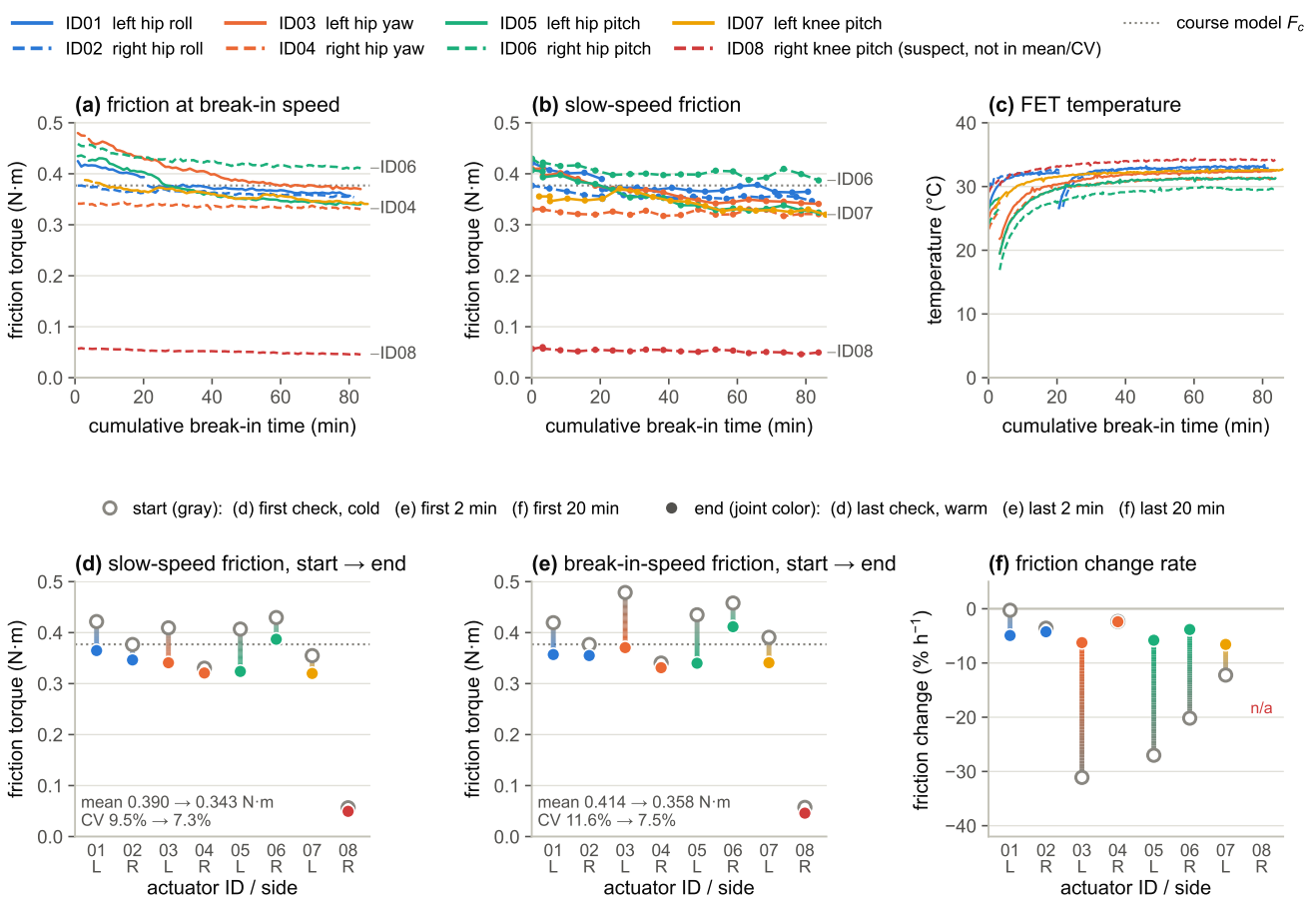

ID01 left hip roll
ID02 right hip roll
ID03 left hip yaw
ID04 right hip yaw
ID05 left hip pitch
ID06 right hip pitch
ID07 left knee pitch
ID08 right knee pitch (suspect, not in mean/CV)
course model F_c
(a) friction at break-in speed
friction torque (N·m)
cumulative break-in time (min)
ID06
ID04
ID08
(b) slow-speed friction
friction torque (N·m)
cumulative break-in time (min)
ID06
ID07
ID08
(c) FET temperature
temperature (°C)
cumulative break-in time (min)
start (gray): (d) first check, cold (e) first 2 min (f) first 20 min
end (joint color): (d) last check, warm (e) last 2 min (f) last 20 min
(d) slow-speed friction, start → end
friction torque (N·m)
mean 0.390 → 0.343 N·m
CV 9.5% → 7.3%
01 L
02 R
03 L
04 R
05 L
06 R
07 L
08 R
actuator ID / side
(e) break-in-speed friction, start → end
friction torque (N·m)
mean 0.414 → 0.358 N·m
CV 11.6% → 7.5%
actuator ID / side
(f) friction change rate
friction change (% h⁻¹)
n/a
actuator ID / side